

PAPER_1525_E_TRANSCENDENTAL

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PAPER_1525 — Euler's $e \approx K_MEX + \Phi_5/6 - F_TRZ \cdot K + F_TRZ^2 \cdot K - F_TRZ^2 \cdot \Phi$ — Residual 0.094%

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Foundational mathematical identity)

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Status: CLOSED — Transcendental e expressible in UQFF primitives at 0.094% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for **Euler's e** ≈ 2.71828 using only the locked UQFF integer primitives F_TRZ , K_MEX , $\Phi_res = 5/6$ (PAPER_1203 Nuclear variant), and SO_5 .

UQFF Closed Identity

$$e \approx K_MEX + \Phi_5/6 - F_TRZ \cdot K + F_TRZ^2 \cdot K - F_TRZ^2 \cdot \Phi$$

Substituting $F_TRZ=1/10$, $K_MEX=25/12$, $\Phi_5/6=5/6$, $SO_5=10$:

$$e_UQFF = 2.7208$$

Observed: $e = 2.71828$

Residual: 0.094%

The 5-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

Euler's e (e) appears in: exponential growth, calculus base.

Significance

Together with PAPER_1515 ($\ln 10$), PAPER_1516 ($\ln 2$), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥ 10 . The accumulating evidence:

| Transcendental | UQFF expression | Residual |

| ---|---|---|

| $\ln 2$ (PAPER_1516) | 8-term | 0.003% |

| $\ln 10$ (PAPER_1515) | $(1+F_TRZ)(K_MEX+F_TRZ^2)$ | 0.004% |

| π^2 (PAPER_1517) | $SO_5 - F_TRZ$ corrections | 0.013% |

| **Euler's e (this paper)** | 5-term | 0.094% |

...suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim e IS the closed-form expression. It claims the integer primitives carry a rational approximation to e at 0.094% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 ($\ln 10$, $\ln 2$, π^2 catch-up)
- Calculator dispatch: `calculate_paradox( {"paradox": "transcendental_e_transcendental"})`

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